Bioinformatics algorithms

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Genome assembly – exercises (**with solutions**)

Exercise 1.

Recall our definitions of our idealized genome from the lecture.

- G: genome size
- L: read length
- N: read number
- n_b: total number of total bases (analogous for k-mers)
- d_b: mean base depth (analogous for k-mers)

Let G = 5 Megabases (approximately the genome of *E.coli*), L = 200 bases

(Note: *Escherichia coli* is a bacterion which is frequently used as a "model organism", especially for basic biochemical functions and pathways.)

In the lecture, we calculated the sequencing depth needed such that at least 99% of the human genome would be covered by at least one read. Under the same slightly unrealistic assumptions as in the lecture, calculate the number of 200 bp reads required to ensure that at least 95% of the *E. coli* genome is covered.

Solution
If we want to estimate the mean read depth required such that at least 95% of the genome is covered once (and thus the probability of any base is at least 95%), we have

$$\begin{aligned} 0.95 = P(X > 0) &= 1 - P(X = 0) \\ 0.95 &= 1 - \frac{e^{-\lambda}\lambda^0}{0!} \\ 0.95 &= 1 - e^{-\lambda} \\ e^{-\lambda} &= 0.05 \\ -\lambda &= -2.995732 \\ \lambda &= 2.995732 \end{aligned}$$

We saw that λ, the mean depth per base in the genome, is equivalent to the total number of sequenced bases n_b divided by the genome size G. The total number of sequenced bases (n_b) is equal to the number of reads N times the read length: $n_b = N \times L$

$$\lambda = d_b = \frac{n_b}{G} = \frac{N \times L}{G} \tag{1}$$

Therefore,

$$\begin{aligned} \lambda &= \frac{N \times L}{G} \\ N &= \frac{\lambda \cdot G}{L} \\ &= \frac{2.995732 \cdot 5 \times 10^6}{200} \\ &= 74,893 \approx 75,000 \end{aligned}$$

Exercise 2.

Refering to the formulas from the lecture, explain why demanding a larger overlap between reads in order to combine reads into contigs is associated with a lower expected contig size.

Solution
Consider an interval I that is as long as a read (L nucleotides). The expected number of reads that start in I is then $\lambda = L \times N/G$. Assuming a Poisson distribution, the probability that no read starts in I is then

$$P(X = 0) = \frac{e^{-\lambda}\lambda^0}{0!} = e^{-\lambda} \tag{2}$$

We have seen that the expected size of the sequenced region of the genome is $G{\cdot}(1 - e^{-\lambda})$. The expected number of contigs is $Ne^{-\lambda}$. Therefore, the expected size of a contig is simply

$$\frac{G \cdot (1 - e^{-\lambda})}{Ne^{-\lambda}} \tag{3}$$

If we demand a certain overlap, the number of sequenced bases remains fixed, but the expected number of contigs needs to be adjusted: Lambda needs to be adjusted to reflect the expected number of reads that start in $(1 - \theta)L$ to be $(1 - \theta)\lambda$.

$$\frac{G \cdot (1 - e^{-\lambda})}{Ne^{-(1-\theta)\lambda}} \tag{4}$$

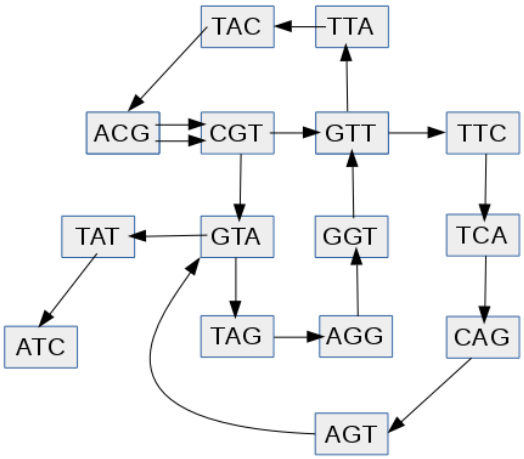
Therefore: the higher θ, the higher the number of contigs and the smaller the expected contig size!

Exercise 3.

a) Construct a de Bruijn graph for the following short DNA sequence, using a decomposition of the sequence into 4-mers:

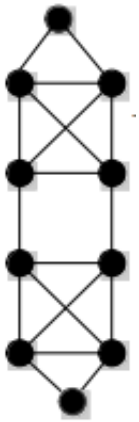
`A C G T T A C G T A G G T T C A G T A T C`

Solution
We need 4-mer edges, hence we use 3-mer nodes!



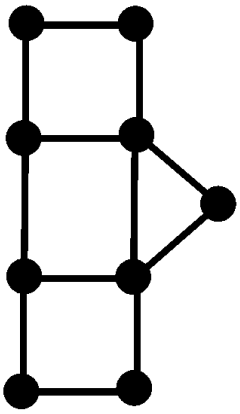
Exercise 4.

a) Does the following graph have a Eulerian path? Does it have a Eulerian cycle?



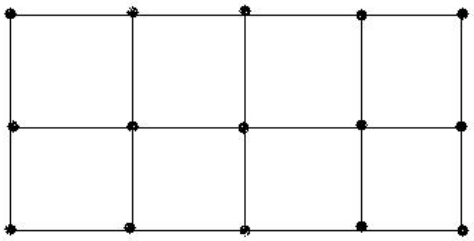
Solution
Yes, all vertices have even degree (Eulerian cycle!). Therefore you can even start at ANY vertex.

b) Does the following graph have a Eulerian path? Does it have a Eulerian cycle?



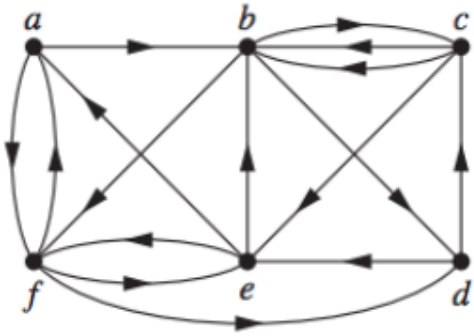
Solution
Yes, as long as you start at one of the two vertices with an odd degree, you can find an Eulerian path (but no cycle).

c) Does the following graph have a Eulerian path? Does it have a Eulerian cycle?



Solution
There are eight vertices of odd degree, therefore the graph doesn't allow an Eulerian path or cycle.

d) Does the following graph have a Eulerian path? Does it have a Eulerian cycle?



Solution
Vertex b has one incoming edge more than outgoing edges; vertex c has one more outgoing edges than incoming edges; all other vertices have the same number of incoming and outgoing edges, therefore the graph has an Eulerian path but not a cycle. The starting node is c!

e) Take one of the three undirected graphs (a-c) which has a Eulerian path but not a Eulerian cycle.

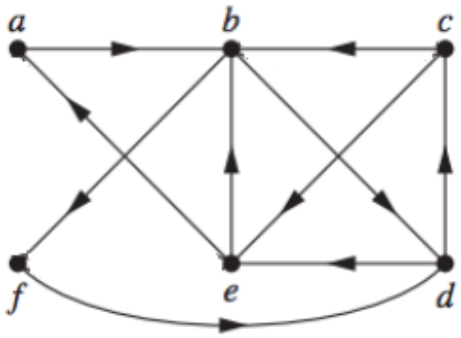
- Can you slightly modify the graph such that it has a Eulerian cycle? If so, how?
- What if the graph was a directed graph with a Eulerian path, could you still modify it to become a Eulerian cycle? If so, how?

Solution
Only graph b) has a Eulerian path but not a cycle.

Easy: add an additional edge to connect the two nodes with odd degree. Now they have even degree like all other nodes; so we have a Eulerian cycle!

In case of a directed graph, the additional edge would need to point from the end node to the start node.

Exercise 5.



For the directed graph shown above, use Hierholzer's approach to identify a Eulerian path!

Solution

- Starting node: c; end node: b
- Eulerian paths:
  1. `epath = [c e b d c b f d e a b]`
  2. `epath = [c b d c e b f d e a b]`
  3. `epath = [c e a b f d e b d c b]`
  4. ... (?)